Def. Let R be a Commutative Ring with $1 \neq 0$.
The polynomial ring in the indeterminate x with coefficients from R is;

$$R[x] \overset{\text{def}}{=} \bigcup_{n=0}^{\infty} \prod_{k=0}^{\infty} X_k^n \quad \text{where} \quad X_k^n = \begin{cases} R & \text{if } k \leq n \\ \{0\} & \text{if } k > n. \end{cases}$$

Denote a tuple $(a_0, a_1, \cdots, a_n, 0, 0, \cdots) \in \prod_{k=0}^{\infty} X_k^n$ as;

$$a_0 + a_1 x + \cdots + a_n x^n$$

And, the addition is given by entry-wise, the multiplication is given by

$$\left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{i=0}^{m} b_i x^i\right) = \sum_{k=0}^{n+m} \underbrace{\left(\sum_{i=0}^{k} a_i b_{k-i}\right)}_{\text{Cauchy Product}} \cdot x^k$$

Prop. Let $I \subset R$ be an Ideal of R. Then, in $R[x]$,

$$(I) = I[x]$$

Moreover

$$R[x]/(I) = R[x]/I[x] \cong (R/I)[x].$$

Proof. $(I) = I \cdot (R[x]) = (IR)[x] = I[x]$, Since R is Commutative, and has 1.
Now, define a map $\varphi : R[x] \to (R/I)[x] : \sum a_n x^n \mapsto \sum (a_n + I) x^n$

Then, $\ker \varphi = I[x]$, so proved by the first isomorphism Theorem.
(More Specifically,

$$\sum a_n x^n \in \ker \varphi \iff \sum (a_n + I) x^n = 0$$
$$\iff \text{all coefficients } a_n \text{ are contained in } I$$
$$\iff \sum a_n x^n \in I[x].$$

「Thm. If R is Noetherian Ring, then $R[x]$ is Noetherian.
Proof Let an ideal $I \subseteq R[x]$ be given, and let $L \subseteq R$ be the set of all leading coefficient of the elements in I. Claim: L is ideal of R.
Since $0 \in I$, so $0 \in L$. Let $a, b \in L$ and $r \in R$ be given.
Then, there are two polynomials in I,

$$f(x) = ax^d + \cdots \quad \text{and} \quad g(x) = bx^e + \cdots$$

Since I is Ideal,

$r \cdot x^e \cdot f(x) - x^d \cdot g(x)$ is contained in I, so $ra - b \in L$.

Which shows $L \subset R$ is an Ideal. Now, L is finitely generated, by assumption for R.
Denote $L = (a_1, \cdots, a_n)$ where $a_1, \cdots, a_n \in R$.

And, Put $f_i \in I$ be a polynomial of degree e_i with the leading coefficient is a_i.
Let $N = \max\{e_1, \cdots, e_n\}$. For each $d \in \{0, 1, \cdots, N-1\}$, Construct L_d be the set of all leading coefficients of polynomial in I of degree d, with 0.
Similarly, each L_d is Ideal of R, so these are generated finitely:

$$L_d = (b_{d1}, \cdots, b_{d,n_d}) \text{ for some } b_{d,i} \in R.$$

Now, for $f_{d,i}$, the polynomial of degree d with the leading coefficient is $b_{d,i}$, the set $\{f_1, \cdots, f_n\} \cup \{f_{di} \mid 0 \le d < N, 1 \le i \le n_d\}$ generates I, because:

Clearly the generated ideal is contained in I. If there is a polynomial $f(x) \in I$ which is not contained in the generated ideal with the minimal degree.
If $\deg f \ge N$, then the leading coefficient a of f can be represent as:

$$a = r_1 a_1 + \cdots + r_n a_n \quad (r_i \in R), \ R\text{-linear Combination of generators for } I.$$

Then, $g = r_1 x^{d-e_1} f_1 + \cdots + r_n x^{d-e_n} f_n$ has a degree smaller than $\deg f$, with the leading coefficient is a, Contradiction to the minimality.
If $\deg f < N$, $a \in L_d$ $(d < N)$, so contradiction.

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